Software Documentation

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Section: C

1. Software Overview

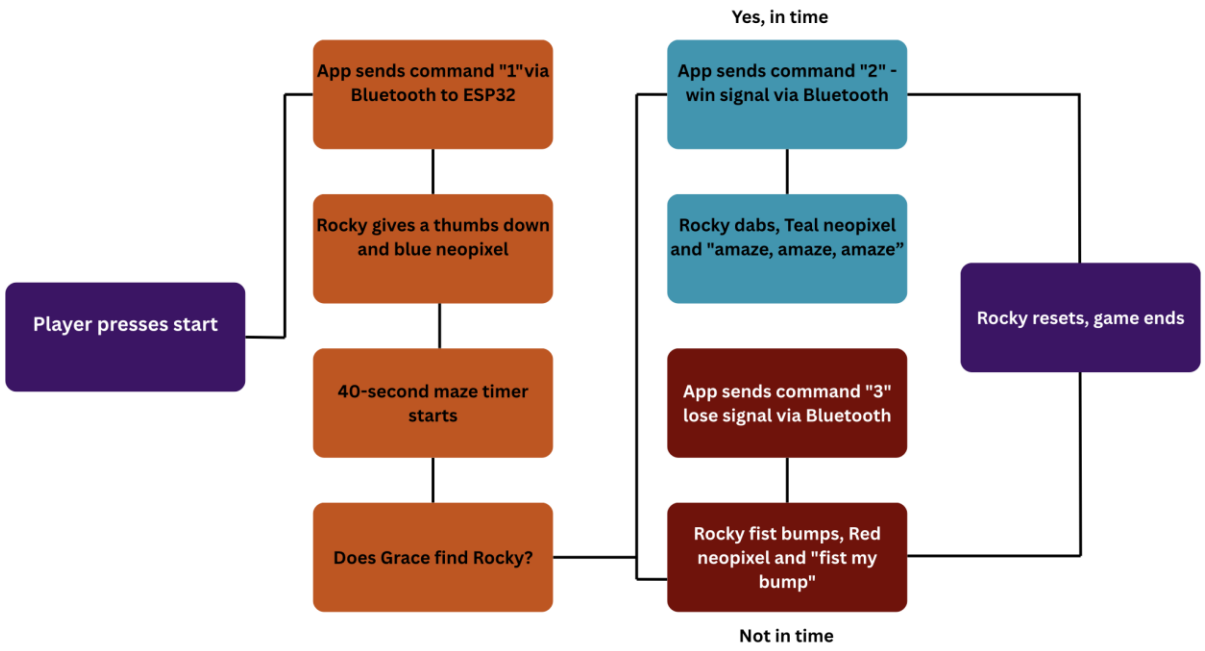
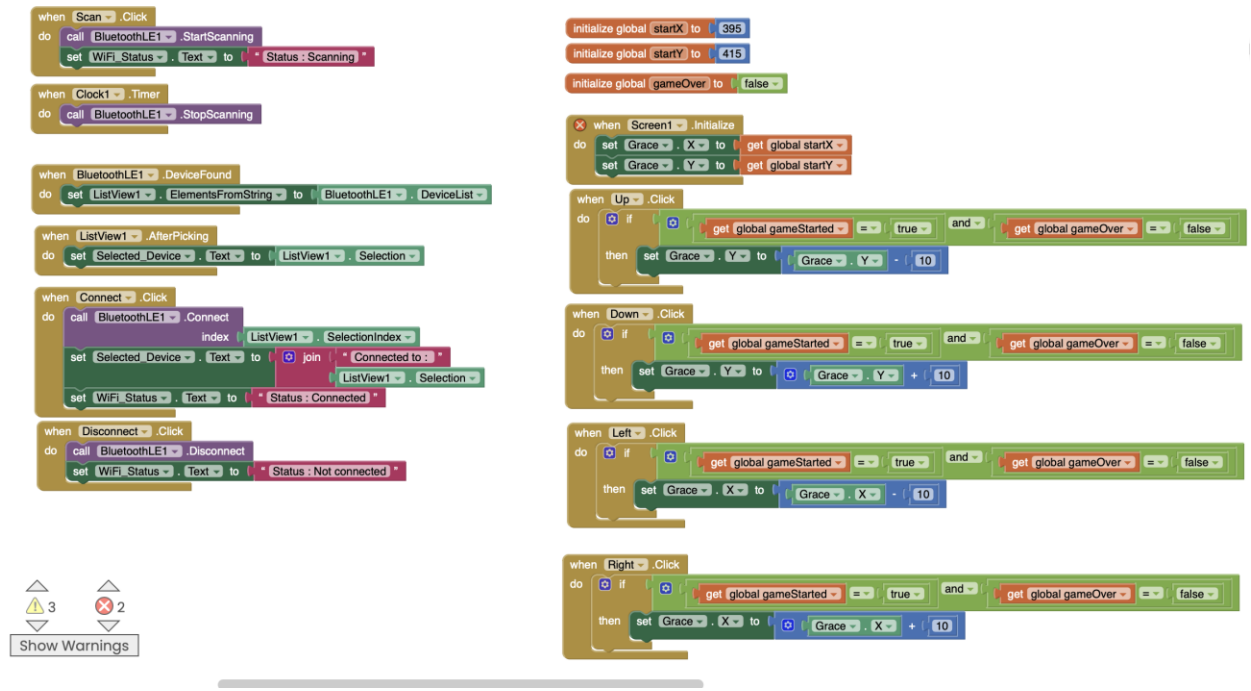
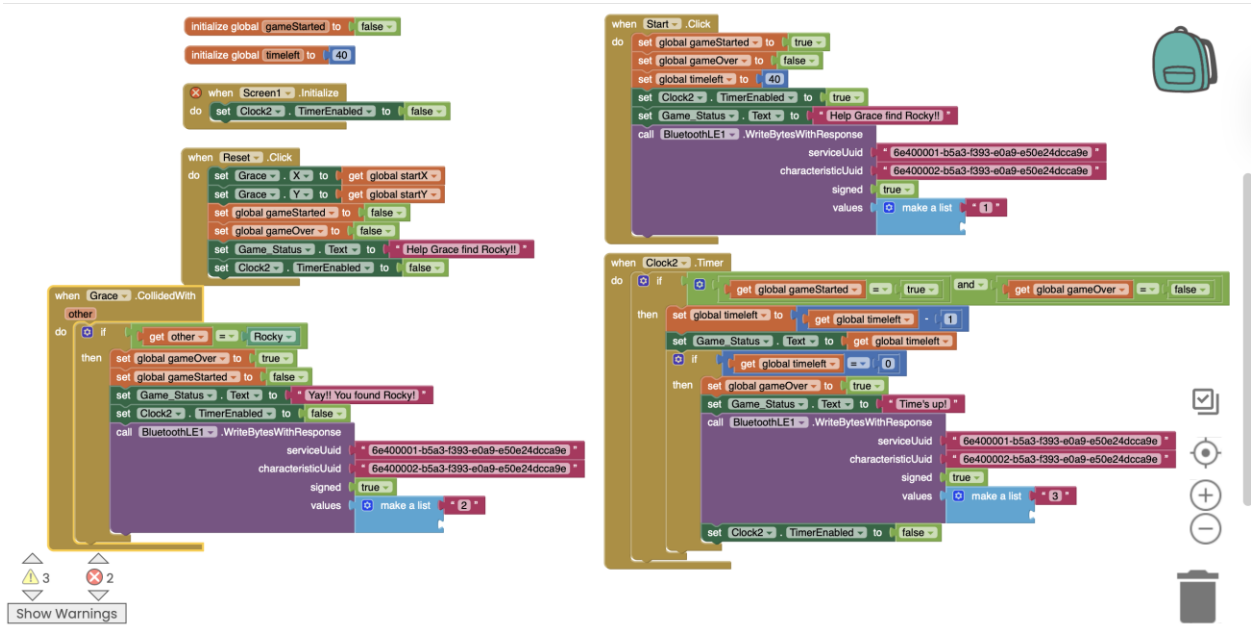
The software connects a mobile game to a physical robot. The MIT App Inventor app serves as the game interface. the player navigates Grace through a maze with a 40-second timer to find Rocky. Based on the outcome, the app sends a Bluetooth signal to the ESP32 microcontroller inside robotic Rocky. The ESP32 runs MicroPython, which interprets the signal and triggers the correct physical response, moving Rocky's servo motors into a gesture and lighting up the NeoPixel LED ring.

Platforms used: MIT App Inventor (mobile game), MicroPython on ESP32 (robot control), BLE (communication between app and robot).

2. System Logic

- Player opens the app and presses Start
- App sends command "1" via Bluetooth to the ESP32
- ESP32 receives "1" - neopixel turns blue → Rocky performs thumbs up (good luck)
- The 40-second maze timer begins - player navigates Grace through the maze
- If Grace finds Rocky in time, app sends command "2" - neopixel turns teal green - Rocky dabs and says "amaze, amaze, amaze"
- If the timer runs out, app sends command "3" - neopixel turns red - Rocky gives a consolation fist bump and says "fist my bump"
- Rocky resets and waits for the next game

3. Flowchart


Yes, in time
App sends command "1"via Bluetooth to ESP32
App sends command "2" - win signal via Bluetooth
Rocky gives a thumbs down and blue neopixel
Rocky dabs, Teal neopixel and "amaze, amaze, amaze"
Player presses start
Rocky resets, game ends
40-second maze timer starts
App sends command "3" lose signal via Bluetooth
Does Grace find Rocky?
Rocky fist bumps, Red neopixel and "fist my bump"
Not in time

4. App Flow (If MIT App Inventor)